1 Data

The estimation dataset is constructed in STATA directly from the raw source files. Each source is first converted into a quarterly dataset indexed by the STATA quarterly date, after which the component datasets are merged onto a contiguous quarterly time spine.

Python is not used for the general data build. The only exception is the PCHIP interpolation of the annual competition series, since STATA does not provide a native monotone-cubic PCHIP routine. All other transformations, filters, merges, lags, and final data exports are carried out in STATA.

1.1 Inflation

Source. Headline CPI, PPI, core CPI, core PCE, and headline PCE are obtained from **FRED**.

Series. All inflation measures are constructed from monthly price indexes. Monthly observations are first averaged within calendar quarter.

Construction. For each quarterly price index P_t, year-on-year inflation is defined as the four-quarter log difference,

$$\pi_t^{\text{yoy}} = 100\left(\log P_t - \log P_{t-4}\right). \tag{1}$$

The corresponding quarter-on-quarter inflation rate is

$$\pi_t^{\text{qoq}} = 100\left(\log P_t - \log P_{t-1}\right). \tag{2}$$

The same construction is applied to headline CPI, PPI, core CPI, core PCE, and headline PCE. For quarterly NKPC specifications, the quarter-on-quarter measure is the natural counterpart to the one-period inflation rate in the model. Annualized quarter-on-quarter inflation can equivalently be expressed as

$$\pi_t^{\text{qoq,ann}} = 400\left(\log P_t - \log P_{t-1}\right). \tag{3}$$

1.2 Inflation Expectations

We retain inflation expectations at their documented forecast horizons. The one-year-ahead series are used as broader short-run expectation measures, while the SPF one-quarter-ahead GDP price-index forecast provides the expectation measure that maps most directly into a quarterly NKPC.

1.2.1 Cleveland Fed Inflation Expectations

Source. Inflation expectations are obtained from the Cleveland Fed inflation-expectation series.

Series. The monthly one-year-ahead expectation series is converted to percentage units and averaged within calendar quarter.

Construction. The Cleveland Fed expectation is retained at its original one-year horizon. We do not divide the one-year-ahead rate by four to construct a one-quarter-ahead forecast, because such a transformation would not in general recover $E_t\pi_{t+1}$.

1.2.2 Survey of Professional Forecasters

Source. Inflation expectations are also obtained from the **Survey of Professional Forecasters (SPF)** of the Federal Reserve Bank of Philadelphia.

One-year-ahead forecasts. The SPF one-year-ahead file reports GDP-price-index and CPI inflation expectations. These are median-based expectations for average inflation over the four quarters following the quarter in which the survey is conducted. The GDP-price-index series is constructed from the median level forecasts for the survey quarter and four quarters ahead, while the CPI series is constructed from the quarterly CPI inflation forecasts over the following four quarters. The published one-year-ahead series are retained directly, with no frequency conversion or interpolation.

One-quarter-ahead GDP price-index forecast. For the quarterly NKPC, we additionally construct the SPF one-quarter-ahead GDP price-index inflation forecast from the *mean* level forecasts. In the SPF level-file nomenclature, PGDP2 denotes the mean forecast of the GDP price-index level for the quarter in which the survey is conducted, while PGDP3 denotes the mean forecast for the following quarter. The SPF official mean growth rate is the growth rate of these mean level forecasts; it is not the cross-sectional mean of individual growth-rate forecasts.

Following the SPF quarterly-growth convention, the one-quarter-ahead forecast is therefore

$$\widehat{\pi}_{t+1|t}^{\mathrm{SPF},1q} = 100\left[\left(\frac{\widehat{P}_{t+1|t}^{\mathrm{mean}}}{\widehat{P}_{t|t}^{\mathrm{mean}}}\right)^4 - 1\right], \tag{4}$$

where the SPF expresses quarterly growth in annualized percentage points using discrete compounding.

Because the NKPC inflation measures in this dataset are also retained in annualized log-difference form, we additionally construct the log-equivalent expectation

$$\widehat{\pi}_{t+1|t}^{\mathrm{SPF},1q,\log} = 400\left(\log \widehat{P}_{t+1|t}^{\mathrm{mean}} - \log \widehat{P}_{t|t}^{\mathrm{mean}}\right). \tag{5}$$

The first measure follows the SPF official quarterly-growth convention; the second is a transformation into the continuous-compounding units used by the annualized log-inflation specifications and is not itself an official SPF reported growth series.

1.3 Output Gap

We consider several alternative measures of real marginal cost and economic slack that can enter the Phillips curve forcing term.

1.3.1 Beveridge–Nelson Output Gap

Source. The Beveridge–Nelson output measure is constructed from quarterly real GDP.

Series. The decomposition provides a transformed output series together with its cyclical component.

Construction. The corresponding trend is reconstructed as

$$y_t^{\mathrm{BN,trend}} = y_t^{\mathrm{BN}} - y_t^{\mathrm{BN,gap}}. \tag{6}$$

1.3.2 Hodrick–Prescott Output Gap

Source. The HP output gap is constructed from quarterly real GDP.

Series. Real output is transformed into 100 log points before filtering.

Construction. Let Y_t denote real output. The transformed series is

$$y_t = 100 \log Y_t. \tag{7}$$

The quarterly HP filter uses the conventional smoothing parameter

$$\lambda = 1600. \tag{8}$$

The decomposition is

$$y_t = \tau_t^{\mathrm{HP}} + c_t^{\mathrm{HP}}, \tag{9}$$

where τ_t^{HP} denotes the trend and
c_t^{HP} denotes the cyclical output gap.

1.3.3 Inverse Markup

Source. The business markup series is taken from Nekarda and Ramey.

Series. Quarterly observations are available directly from the source. Because the markup exhibits substantial low-frequency variation, we also use a Beveridge–Nelson decomposition to isolate its cyclical component.

Construction. The inverse markup is defined as

$$x_t = \frac{1}{\mu_t}, \tag{10}$$

where μ_t denotes the markup.
The cyclical inverse-markup measure is obtained from the Beveridge–Nelson decomposition and provides an alternative empirical proxy for real marginal cost.

1.3.4 Labor-Share Gap

Source. The labor-share measure is obtained from the seasonally adjusted labor-share index available from **FRED**.

Series. The quarterly labor-share index is restricted to positive observations.

Construction. Let L_t denote the labor-share index. The transformed series is

$$\ell_t = 100 \log L_t. \tag{11}$$

The quarterly HP filter uses

$$\lambda = 1600, \tag{12}$$

and decomposes the series as

$$\ell_t = \tau_{L,t}^{\mathrm{HP}} + c_{L,t}^{\mathrm{HP}}, \tag{13}$$

where $\tau_{L,t}^{\mathrm{HP}}$ denotes the labor-share trend and $c_{L,t}^{\mathrm{HP}}$ denotes the cyclical labor-share gap.

1.3.5 Negative Unemployment Gap

Source. The natural rate of unemployment and the seasonally adjusted unemployment rate are obtained from **FRED**.

Series. Both monthly series are averaged within calendar quarter.

Construction. The negative unemployment gap is defined as

$$u_t^{\mathrm{gap}} = u_t^* - u_t, \tag{14}$$

where u_t^* denotes the natural rate of unemployment and u_t denotes the observed unemployment rate.
A positive value therefore corresponds to unemployment being below its estimated natural rate.

1.4 Competition Measures

We use three measures of product-market competition constructed from distinct data sources. The first is based on the industry-concentration data of Grullon, Larkin, and Michaely (2019). The second is based on the Hoberg–Phillips Text-based Network Industry Classifications (TNIC). The third is constructed directly from quarterly firm-level revenue data from S&P Capital IQ.

1.4.1 Grullon–Larkin–Michaely Measure

Source. The first competition measure is based on the industry-concentration data associated with Grullon, Larkin, and Michaely (2019),
Are US Industries Becoming More Concentrated?

Series. The original measure is observed at annual frequency. We retain the level series together with its Beveridge–Nelson cycle and trend components.

Construction. For quarterly-frequency specifications, the annual series and its Beveridge–Nelson components are interpolated to quarterly frequency using monotone-cubic PCHIP interpolation.
For mixed-frequency specifications, the annual observations are instead used without interpolation. The observed annual value is assigned to Q4 of the corresponding year, while Q1–Q3 are left missing.
The two representations therefore serve different purposes. The interpolated series provide a quarterly path for quarterly-frequency estimation, while the Q4-only series preserve the original annual information at its native frequency for mixed-frequency estimation.
Because the PCHIP knots are located at year-end, the interpolated and observed-only representations coincide at Q4.

1.4.2 Hoberg–Phillips TNIC Measure

Source. The second competition measure is based on the Hoberg–Phillips Text-based Network Industry Classifications (TNIC). TNIC identifies product-market competitors using pairwise similarity in firms' 10-K product descriptions, allowing competitor sets to vary across firms.

Series. The resulting competition measure is observed at annual frequency. We retain the level series together with its Beveridge–Nelson cycle and trend components.

Construction. For quarterly-frequency specifications, the annual series and its Beveridge–Nelson components are interpolated to quarterly frequency using monotone-cubic PCHIP interpolation.
For mixed-frequency specifications, no interpolation is applied. The original annual observations are retained at Q4, while Q1–Q3 are left missing.
As with the Grullon–Larkin–Michaely measure, the interpolated series is used for quarterly-frequency estimation and the observed-only series preserves the native annual observations for mixed-frequency estimation. The two representations coincide at Q4.

1.4.3 Capital IQ Competition Measure

Source. The third competition measure is constructed directly from S&P Capital IQ. The sample consists of U.S. public firms listed on major exchanges, with firm-level total revenue observed at quarterly frequency.

Series. Unlike the two annual competition measures, the Capital IQ measure is constructed natively at quarterly frequency and therefore requires no temporal interpolation.

Firm-quarter observations with missing reported revenue are excluded. Negative reported revenues are floored at zero when constructing market shares, while the original reported values are retained separately.

Industry classification is based on SIC. A firm's current SIC classification is applied to its full revenue history.

Market definition. The baseline broad-market definition is based on SIC divisions, with manufacturing separated into durable and nondurable goods so that the entire manufacturing sector is not treated as a single market.

Alternative market definitions based on SIC, NAICS, and S&P sector classifications are also retained in the underlying competition panel.

Firm revenue shares. Within market j and quarter t, firm i's revenue share is

$$s_{ijt} = \frac{R_{ijt}}{\sum_{i \in j} R_{ijt}}, \tag{15}$$

where R_{ijt} denotes cleaned quarterly revenue.

Market concentration. Market-level concentration is measured by the Herfindahl–Hirschman index,

$$HHI_{jt} = \sum_{i \in j} s_{ijt}^2. \tag{16}$$

The baseline construction retains only market-quarter observations containing at least 10 firms.

Seasonal adjustment. Each market-level HHI is seasonally adjusted before economy-wide aggregation. The adjustment removes systematic within-year movements in concentration within each market while retaining idiosyncratic quarter-specific variation. Let HHI_{jt}^{SA} denote the resulting seasonally adjusted market concentration measure.

Economy-wide effective number of firms. The economy-wide effective number of firms is defined as the inverse of the weighted mean market HHI,

$$N_t = \frac{1}{\sum_j w_{jt} HHI_{jt}^{\mathrm{SA}}}, \qquad \sum_j w_{jt} = 1. \tag{17}$$

We construct both firm-count-weighted and revenue-weighted versions,

$$N_t^{\mathrm{firm}} = \frac{1}{\sum_j w_{jt}^{\mathrm{firm}} HHI_{jt}^{\mathrm{SA}}}, \tag{18}$$

and

$$N_t^{\mathrm{rev}} = \frac{1}{\sum_j w_{jt}^{\mathrm{rev}} HHI_{jt}^{\mathrm{SA}}}. \tag{19}$$

Because these measures are constructed directly from quarterly firm-level revenues, no temporal interpolation is required.

1.5 Establishment Stock

Source. The establishment-stock measure combines the annual Business Dynamics Statistics (BDS) establishment count with quarterly Business Employment Dynamics (BED) establishment births and deaths.

Series. The BDS data provide a level benchmark for the number of establishments, while the BED data provide quarterly establishment birth and death flows. The BED series are reported in thousands and are converted to establishment counts before constructing the stock.

Construction. The 1993 BDS establishment count is used as the level anchor. Quarterly establishment net entry is defined as

$$\Delta E_t = B_t - D_t, \tag{20}$$

where B_t denotes establishment births and D_t denotes establishment deaths. The establishment stock is then constructed as

$$E_t = E_{1993}^{\text{BDS}} + \sum_{s \le t} \Delta E_s. \tag{21}$$

1.6 Lags and Timing

For regressors entering the empirical specifications with a one-quarter lag, we also retain the corresponding lagged observation. For a generic quarterly series z_t,

$$z_t^{\text{prev}} = z_{t-1}. \tag{22}$$

Quarterly observations are dated at the end of the corresponding calendar quarter.

1.7 Variable Definitions

Table 1 summarizes the variable names used in the model-ready dataset. Variable names are reported here, rather than in the preceding discussion, to separate the economic definitions from their implementation in the data files.

Table 1: Variables in the Model-Ready Dataset

Category	Variable	Description
Inflation	`pi_cpi`	Headline CPI inflation, year-on-year log difference.
Inflation	`pi_cpi_qoq`	Headline CPI inflation, quarter-on-quarter log difference.
Inflation	`pi_cpi_qoq_ann`	Annualized quarter-on-quarter headline CPI log inflation.
Inflation	`pi_cpi_core`	Core CPI inflation, year-on-year log difference.
Inflation	`pi_cpi_core_qoq`	Core CPI inflation, quarter-on-quarter log difference.
Inflation	`pi_cpi_core_qoq_ann`	Annualized quarter-on-quarter core CPI log inflation.

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Table 1 – continued from previous page

Category	Variable	Description
Inflation	pi_pce	Headline PCE inflation, year-on-year log difference.
Inflation	pi_pce_qoq	Headline PCE inflation, quarter-on-quarter log difference.
Inflation	pi_pce_qoq_ann	Annualized quarter-on-quarter headline PCE log inflation.
Inflation	pi_pce_core	Core PCE inflation, year-on-year log difference.
Inflation	pi_pce_core_qoq	Core PCE inflation, quarter-on-quarter log difference.
Inflation	pi_pce_core_qoq_ann	Annualized quarter-on-quarter core PCE log inflation.
Inflation	pi_ppi	PPI inflation, year-on-year log difference.
Inflation	pi_ppi_qoq	PPI inflation, quarter-on-quarter log difference.
Inflation	pi_ppi_qoq_ann	Annualized quarter-on-quarter PPI log inflation.
Expectations	Epi	Cleveland Fed one-year-ahead inflation expectation, retained at its original horizon.
Expectations	Epi_spf_gdp	SPF one-year-ahead GDP-price inflation expectation; median-based average over the four quarters following the survey quarter.
Expectations	Epi_spf_cpi	SPF one-year-ahead CPI inflation expectation; median-based average over the four quarters following the survey quarter.
Expectations	Epi_spf_gdp_1q	SPF one-quarter-ahead GDP price-index growth constructed from mean level forecasts using the official annualized discrete Q/Q convention.
Expectations	Epi_spf_gdp_1q_log	Annualized log-change equivalent of the SPF one-quarter-ahead mean-level forecast; not an official SPF reported growth series.
Competition	N_Gustavo	PCHIP-interpolated quarterly Grullon–Larkin–Michaely competition series.
Competition	N_Gustavo_BN_cycle	Beveridge–Nelson cycle of the Grullon–Larkin–Michaely series, interpolated quarterly.
Competition	N_Gustavo_BN_trend	Beveridge–Nelson trend of the Grullon–Larkin–Michaely series, interpolated quarterly.
Competition	N_Gustavo_annual_q4	Original annual Grullon–Larkin–Michaely observation retained at Q4 only.
Competition	N_Gustavo_BN_cycle_annual_q4	Original annual Beveridge–Nelson cycle retained at Q4 only.
Competition	N_Gustavo_BN_trend_annual_q4	Original annual Beveridge–Nelson trend retained at Q4 only.
Competition	N_TNIC	PCHIP-interpolated quarterly Hoberg–Phillips TNIC competition series.
Competition	N_TNIC_BN_cycle	Beveridge–Nelson cycle of the TNIC series, interpolated quarterly.
Competition	N_TNIC_BN_trend	Beveridge–Nelson trend of the TNIC series, interpolated quarterly.
Competition	N_TNIC_annual_q4	Original annual TNIC observation retained at Q4 only.
Competition	N_TNIC_BN_cycle_annual_q4	Original annual TNIC Beveridge–Nelson cycle retained at Q4 only.
Competition	N_TNIC_BN_trend_annual_q4	Original annual TNIC Beveridge–Nelson trend retained at Q4 only.
Competition	N_capitaliq_firmw	Capital IQ effective number of firms using firm-count-weighted market HHIs.
Competition	N_capitaliq_revw	Capital IQ effective number of firms using revenue-weighted market HHIs.
Output gap	output_BN	Beveridge–Nelson transformed output series.

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Category	Variable	Description
Output gap	output_gap_BN	Beveridge–Nelson cyclical output gap.
Output gap	output_trend_BN	Beveridge–Nelson output trend.
Output gap	output	Log real output.
Output gap	output_trend_HP	Hodrick–Prescott output trend.
Output gap	output_gap_HP	Hodrick–Prescott cyclical output gap.
Output gap	labor_share	Quarterly labor-share index.
Output gap	labor_share_100log	Labor-share index expressed in 100 log points.
Output gap	labor_share_trend_HP	Hodrick–Prescott labor-share trend.
Output gap	labor_share_gap_HP	Hodrick–Prescott labor-share gap.
Output gap	unemp_gap	Negative unemployment gap, natural rate minus observed unemployment.
Output gap	markup	Nekarda–Ramey business markup.
Output gap	markup_inv	Inverse markup.
Output gap	markup_BN_inv	Beveridge–Nelson cyclical inverse-markup measure.
Establishments	establishment_births	Quarterly establishment births.
Establishments	establishment_deaths	Quarterly establishment deaths.
Establishments	establishment_net_entry	Quarterly establishment births minus deaths.
Establishments	establishment_stock	Establishment stock constructed from the 1993 BDS anchor and cumulative BED net entry.
Lag	pi_cpi_prev	One-quarter lag of headline CPI inflation.
Lag	pi_ppi_prev	One-quarter lag of PPI inflation.
Lag	pi_cpi_core_prev	One-quarter lag of core CPI inflation.
Lag	pi_pce_prev	One-quarter lag of headline PCE inflation.
Lag	pi_pce_core_prev	One-quarter lag of core PCE inflation.
Lag	pi_cpi_qoq_prev	One-quarter lag of quarter-on-quarter headline CPI inflation.
Lag	pi_ppi_qoq_prev	One-quarter lag of quarter-on-quarter PPI inflation.
Lag	pi_cpi_core_qoq_prev	One-quarter lag of quarter-on-quarter core CPI inflation.
Lag	pi_pce_qoq_prev	One-quarter lag of quarter-on-quarter headline PCE inflation.
Lag	pi_pce_core_qoq_prev	One-quarter lag of quarter-on-quarter core PCE inflation.
Lag	pi_cpi_qoq_ann_prev	One-quarter lag of annualized quarter-on-quarter headline CPI log inflation.
Lag	pi_ppi_qoq_ann_prev	One-quarter lag of annualized quarter-on-quarter PPI log inflation.
Lag	pi_cpi_core_qoq_ann_prev	One-quarter lag of annualized quarter-on-quarter core CPI log inflation.
Lag	pi_pce_qoq_ann_prev	One-quarter lag of annualized quarter-on-quarter headline PCE log inflation.
Lag	pi_pce_core_qoq_ann_prev	One-quarter lag of annualized quarter-on-quarter core PCE log inflation.
Lag	unemp_gap_prev	One-quarter lag of the negative unemployment gap.
Lag	markup_inv_prev	One-quarter lag of the inverse markup.
Lag	markup_BN_inv_prev	One-quarter lag of the Beveridge–Nelson cyclical inverse markup.

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Category	Variable	Description
Lag	`output_gap_BN_prev`	One-quarter lag of the Beveridge–Nelson output gap.
Lag	`output_gap_HP_prev`	One-quarter lag of the Hodrick–Prescott output gap.
Lag	`labor_share_gap_HP_prev`	One-quarter lag of the Hodrick–Prescott labor-share gap.

2 Summary of Data Construction

The main data choices can be summarized as follows:

- **Data construction.**

  The model-ready dataset is built in STATA directly from the raw source files. The only exception is the PCHIP interpolation of the annual competition measures, which is performed outside STATA.

- **Inflation.**

  Inflation is constructed from quarterly averages of monthly price indexes. Year-on-year, quarter-on-quarter, and annualized quarter-on-quarter log inflation measures are retained. The quarterly measures correspond most directly to the one-period inflation rate in the quarterly NKPC, while the year-on-year measure is retained as an alternative specification.

- **Inflation expectations.**

  Expectations are retained at their documented horizons rather than being mechanically converted across horizons.

  - Cleveland Fed expectations are retained at the one-year-ahead horizon; no division by four is used to approximate a next-quarter expectation.
  - SPF one-year-ahead GDP-price and CPI expectations are the published median-based measures of average inflation over the four quarters following the survey quarter.
  - The SPF one-quarter-ahead GDP price-index measure is constructed from PGDP2 and PGDP3 in the mean level forecasts. Consistent with the SPF documentation, the official-style measure is the annualized discrete Q/Q growth rate of the mean forecast levels, not the mean of individual growth-rate forecasts.
  - A separate annualized log-change equivalent of the one-quarter- ahead SPF forecast is retained for specifications using annualized log inflation.

- **Phillips-curve forcing variable.**

  Several alternative measures of economic slack or real marginal cost are retained:

  - Beveridge–Nelson output gap;

  - Hodrick–Prescott output gap;
  - inverse markup and its Beveridge–Nelson cyclical component;
  - Hodrick–Prescott labor-share gap; and
  - negative unemployment gap.

  This allows the Phillips-curve results to be evaluated without relying on a single empirical proxy for the theoretical forcing variable.

- **Grullon–Larkin–Michaely competition measure.**
  The original measure is annual. For quarterly-frequency specifications, the annual level and its Beveridge–Nelson components are interpolated using monotone-cubic PCHIP. For mixed-frequency specifications, the original annual observations are retained at Q4 without interpolation.

- **Hoberg–Phillips TNIC competition measure.**
  The TNIC-based measure is also annual and is treated in the same way: PCHIP interpolation is used for quarterly-frequency specifications, while the original Q4 observations are used for mixed-frequency specifications.

- **Capital IQ competition measure.**
  The Capital IQ measure is constructed directly at quarterly frequency from revenues of U.S. public firms listed on major exchanges. Firm-level revenue shares are aggregated into market-level HHIs, which are seasonally adjusted before aggregation. Economy-wide effective-firm measures are then constructed using both firm-count and revenue weights.

- **Establishment dynamics.**
  Quarterly establishment births and deaths from BED are combined with the 1993 BDS establishment count as a level anchor. The resulting establishment stock provides a measure of changes in the population of active establishments that is distinct from the revenue-based concentration measures.

- **Timing and lags.**

  Quarterly observations are dated at calendar-quarter end. One-quarter lags are retained for variables entering lagged specifications.